

Natural and man-made fibres: Physical and mechanical properties

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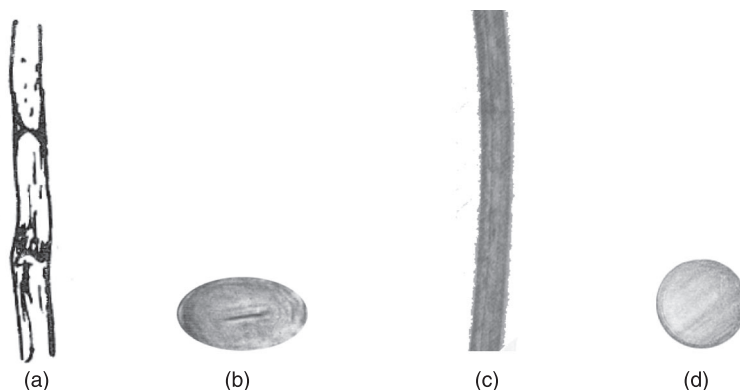
Abstract: A *fibre* is a unit of matter characterized by flexibility, fineness and a high ratio of length to thickness. Because fibres have a high surface to volume ratio, they can be extremely strong materials. According to their origin, textile fibres may be classified as *natural fibres*, when they occur in nature in fibre form, and *man-made fibres*, when they do not occur in nature in fibre form. This chapter addresses the relationship between their structure and properties, and their use in civil engineering applications, such as road construction, bridges, non-structural gratings and claddings, structural systems for industrial supports, buildings, long-span roof structures, tanks, thermal insulators, etc.

Key words: fibres, modulus, strength, density, civil engineering.

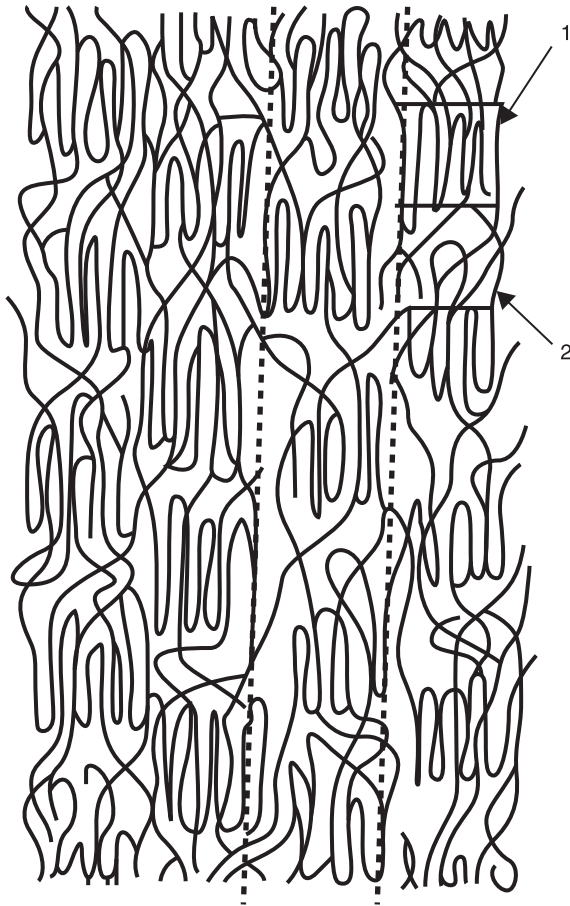
1.1 Introduction

A *fibre* is a unit of matter characterized by flexibility, fineness and a high ratio of length to thickness (Fig. 1.1).¹ As fibres have a high surface to volume ratio, they can be extremely strong materials.

Fibres are normally constituted by long and chain-like molecules known as macromolecules or polymers, which may be of organic or inorganic nature. These molecules are able to pack together closely to each other resulting in regions of



1.1 Typical textile fibres: (a) flax, longitudinal view; (b) flax, cross-sectional view; (c) polyester, longitudinal view; (d) nylon 6.6, cross-sectional view.



1.2 Model of molecular arrangement in nylon fibre:¹² (1) crystalline region; (2) amorphous region.

crystallinity (Fig. 1.2(1)). The degree of orientation of these regions is an important factor in determining the usefulness of a fibre for a particular application. There are other regions, however, where the molecules do not hold together and form random arrangements or amorphous regions (Fig. 1.2(2)). The *crystalline regions* provide strength and rigidity to the fibres, while the *amorphous regions* are responsible for flexibility and reactivity. The ratio of crystalline to amorphous material has an important influence on the properties of the fibres.

According to their origin, textile fibres may be classified as *natural fibres*, when they occur in nature in that form, and *man-made fibres*, when they do not occur in nature in fibre form. Within the latter, some are made of a natural polymer that has been spun into *natural polymer fibres*, and others are made of a synthetic polymer that has been spun into *synthetic fibres*.

High-performance fibres are mostly synthetic fibres that are engineered for specific applications that require very high performance in terms of strength, stiffness, heat resistance or chemical resistance. These fibres have generally higher tenacity and higher modulus than standard fibres.

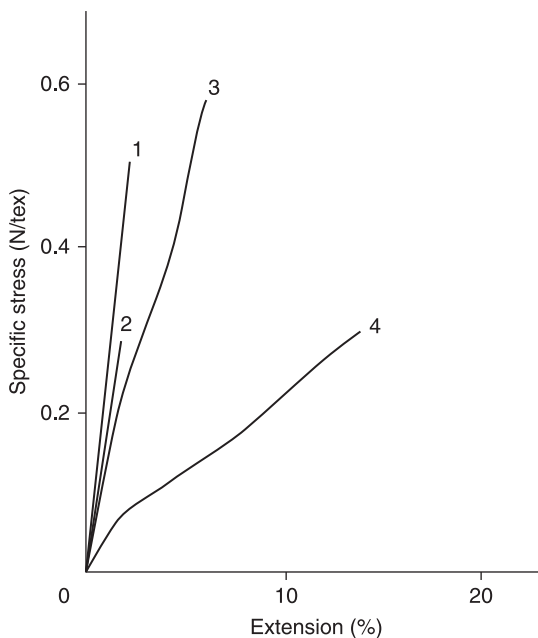
There are also some inorganic *man-made non-polymeric fibres* such as metal threads.

Fibres are used in many civil engineering applications including road construction, bridges, non-structural gratings and claddings, structural systems for industrial supports, buildings, long-span roof structures, tanks, thermal insulators and so on.

1.2 Natural fibres

Traditional natural fibres, such as cotton, wool and silk, have tenacities in the order of 0.1–0.4 N/tex and initial moduli ranging from 2–5 N/tex. However, fibres such as flax, hemp, jute and ramie may have higher strength and stiffness. With the exception of silk, which was used in more demanding applications such as protective and parachute fabrics, they are all short fibres, and this hinders the conversion efficiency of the fibre's strength into yarns and fabrics.

The usefulness of natural fibres in civil engineering applications (Fig. 1.3) is limited by their moderate mechanical properties. However, there may be



1.3 Typical stress–strain curves of cellulosic fibres: (1) flax; (2) jute; (3) fortisan; (4) tenasco.

advantages from using some of these fibres for the reinforcement of composites used as building materials. The low specific gravity of fibres like jute results in a higher specific strength and stiffness than glass fibres and this may be a benefit, especially in parts designed for bending stiffness. The tensile strength and modulus of jute are lower than those of glass fibres; however, the specific modulus of the jute fibre is higher than that of glass and, on a modulus-per-cost basis, jute can also be superior. The specific strength per unit cost of jute, too, approaches that of glass. The advantage of using jute fibres as a substitute for glass fibres, partly or in total, in the reinforcement of composite materials arises from its lower specific gravity (1.50) and higher specific modulus (11.46 N/tex) when compared with glass (2.58 and 10.85 N/tex respectively). Furthermore, the lower cost and the renewable nature of jute, requiring less energy for its production (approximately 2% of that of glass), make it an attractive reinforcing fibre in composites for some civil engineering applications.²

1.3 Man-made fibres

1.3.1 Natural polymer fibres

Viscose rayon and acetate, which were amongst the first regenerated cellulose fibres, exhibited tenacities below 0.2 N/tex. The need to improve the strength of cellulosic fibres led to the development of continuous-filament rayon yarns (*Tenasco*), with a tenacity of 0.4 N/tex, which were used for tyre cords. Further improvements led to strengths of the order of 0.6 N/tex, with an extension to break in the region of 13%. Other developments led to fibres with higher stiffness, such as *Fortisan*, which was developed from highly stretching acetate yarns which were subsequently converted into cellulose. This increased the tenacity to 0.6 N/tex and the modulus to 16 N/tex. Another example is *Bocell*, which is a cellulose fibre spun from a liquid-crystal solution in phosphoric acid and has tenacities and moduli of the order of 1.1 N/tex and 30 N/tex, respectively.³

It should be noted that the extension to break of regenerated fibres is generally high, ranging from around 6–27%. The low to moderate mechanical properties of these fibres render their usefulness quite limited in civil engineering applications at the present time.

1.3.2 Synthetic fibres

The first synthetic fibre was a polyamide, generally known as nylon, which started to be marketed in 1938 and found applications in wartime technical uses. Its tenacity was in the region of 0.5 N/tex and the modulus 2.5 N/tex. Another synthetic fibre that followed was the polyester fibre (polyethylene terephthalate), which had a similar tenacity but a higher modulus of around 10 N/tex. Varieties of these fibres developed for industrial applications, such as ropes and tyre cords,

exhibit tenacities over 0.8 N/tex for both nylon and polyester and moduli of 9 N/tex for nylon and 12 N/tex for polyester.

Another fibre of increasing interest in technical applications is polypropylene. Its tenacity is in the region of 0.65 N/tex with a modulus of 7.1 N/tex and a specific gravity of 0.91.

Nylons, but especially polypropylene and polyester, are widely used in geosynthetics, and polyester is very much used for tensile surface structures, particularly in Europe. Other applications of polypropylene include anti-crack building products for wall surfaces.

High-performance fibres were developed in the latter part of the twentieth century and inhibit a step change in strength and stiffness, compared to previous ones.

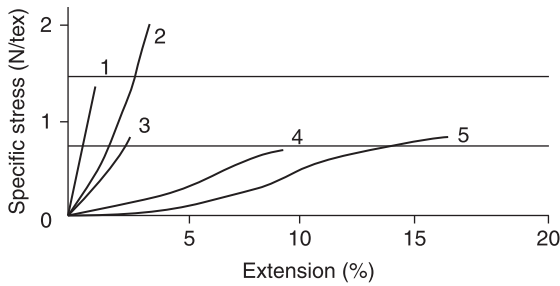
Advances in the 1960s led DuPont scientists in the USA to spin para-aramid fibres from liquid-crystal solutions. These highly oriented fibres achieved tenacities in excess of 2 N/tex and moduli reaching 80 N/tex. More-recent developments enabled some polymer fibres to attain tenacities above 3.5 N/tex and moduli exceeding 150 N/tex.

Other interesting characteristics of para-aramids include flame resistance (self-extinguishing), low electrical conductivity, high chemical resistance, excellent dimensional stability, low thermal shrinkage, high toughness (work of rupture) and high cut resistance. Apart from their use in the reinforcement of composite materials, para-aramids are also of interest for ropes and cables and as a replacement for asbestos materials.

The first high-strength carbon fibres were developed at the Royal Aircraft Establishment in the UK, and were produced by high-temperature processing of acrylic fibres under tension. This resulted in tenacities of up to 3 N/tex and moduli of over 400 N/tex.

Carbon fibre is engineered for strength and stiffness, but variations differ in electrical conductivity, thermal, and chemical properties. The primary factors governing the physical properties are the degree of carbonization, the orientation of the layered carbon planes, and the degree of crystallization. They have lower thermal expansion coefficients than both the glass and aramid fibres, and the material has a very high fatigue and creep resistance.

Much research is also now being done using carbon fibre-reinforced plastic (CFRP) as internal reinforcement in concrete structures, such as beams and bridge decks. The material has many advantages over conventional steel, especially in that it is much stiffer and corrosion resistant. CFRP has become of prominent importance in structural engineering, due to its cost-effectiveness in the repair of old bridges, many of which were designed to tolerate far lower service loads than they are subject to today. Reinforcing with CFRP is a much cheaper alternative when compared with the cost of replacing the bridge. Due to the very high stiffness of CFRP, it can be used underneath spans to help prevent excessive deflections, or wrapped around beams to limit shear stresses.



1.4 Typical stress–strain curves of synthetic fibres: (1) carbon; (2) aramid; (3) glass; (4) polyester; (5) nylon.

Glass fibres are used widely as reinforcements in order to increase the stiffness of composite materials. The tenacity of glass fibres may reach 1.6 N/tex and their moduli 35 N/tex, which are lower than those of aramids (on a weight basis). The development of ceramic fibres is primarily related to their high-temperature performance in metal and ceramic matrix composites to use in engines. However, because of their structure they exhibit high moduli, which may reach 100 N/tex, with tenacities of up to 1 N/tex, which may not be considered particularly high (on a weight basis).

Glass fibres are divided into three classes, i.e. E-glass, S-glass and C-glass. The E-glass is designated for electrical use, and the S-glass for high strength. The C-glass is for high corrosion resistance, and it is not of common use in civil engineering applications. Of the three fibres, the E-glass is the most common reinforcement material used in civil engineering structures.

Glass fibres are also used for fire-protective walls, floors and ceiling panels, as well as fireproof curtains and partitions for indoors and outdoors. They are used for heat insulation in heating systems, power generation and incinerators.²

Basalt fibres have similar applications to carbon and glass fibres in the reinforcement of composite materials, having better physical-mechanical properties than glass and being much cheaper than carbon fibre (Fig. 1.4). Basalt fibre products include: chemically resistant chopped strands for concrete reinforcement; high-strength rovings for pultruded load-bearing parts and concrete reinforcing bars; basalt woven fabrics for heat, sound insulation, and fire protection; geogrids for road and land reinforcement; and stucco nets for wall reinforcing and renovation.

1.4 Textile fibres for use in civil engineering applications: an overview

Most of the fibres used in civil engineering applications are synthetic fibres. In fibre properties evaluation for particular applications, four aspects are of interest,

Table 1.1 Specific properties of some high-performance fibres in comparison with steel

Specific properties	Carbon HM	Para-aramid HM	E-glass	Basalt CBF	Steel
Specific modulus: E/ρ (N.m/kg)	256	80	28	32	27
Specific strength: σ/ρ (N.m/kg)*	1.2	2	0.775	0.75	0.25

ρ = density

σ = tensile strength

E = Young's modulus

*Also known as strength-to-weight ratio

namely: fibre geometry and dimension, mechanical properties, physical properties, and chemical durability (Table 1.1).

A synthetic fibre can be described as a flexible, macroscopically homogeneous body, with a high length-to-thickness ratio (aspect ratio) and a small cross-section. All fibres have one particular structural feature in common: a preferential orientation of their elemental units with respect to the fibre axis.

Synthetic fibres are available in many different geometries and dimensions, such as monofilaments, rovings (tows) and fibrillated polymers.

Monofilaments are single straight fibres usually made by drawing molten polymers. Such fibres are usually round in cross-section, ranging in diameter from 50 μm to 0.5 mm, being usually strong.⁴ However, the smooth surface may result in a low interfacial bond strength, which limits the reinforcing capability of the fibre. Bond strength is primarily determined by the nature of the polymer. If bond strength at the interface between the fibre and the matrix is too high, the fibre ruptures after the first crack initiates. On the other hand, the fibre is easily pulled out if the bond strength is too low. Bond strength may dominate the mechanical properties of fibre-reinforced concrete in this way and the combination of fibre and matrix needs to be selected prudently to get efficient bond strength.

Rovings of polyaramid present the highest bond strength, slightly higher than polyacrylic monofilaments. Polyamide and polyester monofilaments present one of the lowest bond strengths.

Many polymers are spun in the form of filaments (fibrils) whose diameters vary from 5–15 μm . Several thousand fibrils are spun simultaneously, by drawing them from a single die and bundling together. Rovings are cut into short lengths when adding them to concrete. In the concrete, the yarn separates into individual fibrils that need to be dispersed efficiently during mixing. Dispersed rovings act as individual monofilaments within the cement matrix. The high surface-to-volume ratio should increase the effective interfacial bonding.

The geometry of a fibrillated fibre can be described in terms of a film thickness (generally 15–100 μm) and the width of each fibril (100–600 μm). During mixing, the bundles are opened up by the aggregate to act as individual fibrils, each one



1.5 Typical continuous fibre-mat, high-performance, fibre-reinforced cement composite.²

contributing to the reinforcing action. The fibres are easy to handle and to disperse uniformly within a concrete mix (Fig. 1.5).

Fibre-reinforced composites are gaining popularity within the civil construction sector. Applications like fibre-reinforced concrete, concrete retrofitting, concrete jacketing and internal and external reinforcement of composite concrete structures are, among others, applications where fibres and fibrous structures play an important role. In these emerging applications, fibres such as glass, carbon and aramid are of paramount importance because of their high strength, high modulus, and the possibility of competing with the conventional, structural construction materials.

In applications related to the geotechnical and geoenvironmental applications, where properties beside mechanical ones are of paramount importance, fibres and textile structures are already commonly used materials. Examples such as embankments and retaining walls, road construction and road overlaying, erosion control of steepened slopes, shore protection, construction of waste landfills, architectural membranes and offshore applications, among others, show that fibres and fibrous structures are the core construction materials. In these applications, polyamide fibres, and especially polypropylene and polyester fibres, are used in an indiscriminate way.⁴

The use of natural fibres in civil engineering applications is constrained by their moderate mechanical properties. However, their low specific gravity and their

renewable nature make some of these fibres, more specifically the bast fibres, attractive as the reinforcing elements in composites used for some civil engineering applications.

The bast or stem fibres are cellulosic fibres that form the fibrous bundles in the inner bark of the stems of dicotyledenous plants, i.e. plants that form two seed leaves. These fibres are constructed of long thick-walled cells that overlap one another. They are cemented together by non-cellulosic materials to form continuous strands that may run the entire length of the plant stem. The strands of bast fibres are normally released from the cellular and woody tissue of the stem by a process called retting. Often the strands are used commercially without separating the individual fibres one from another. Some examples of bast fibres are flax, hemp, jute and ramie (Table 1.2).

Table 1.2 Tensile properties and density of typical textile fibres used for technical applications in comparison with metals

Fibre	Tenacity (N/tex)	Tensile strength (GPa)	Extension to break (%)	Young's modulus (N/tex)	Young's modulus (GPa)	Density (g/cm ³)
<i>Natural</i>						
Flax	0.54	0.831	3	18	27.72	1.54
Hemp	0.47	0.705	2.2	21.7	32.55	1.50
Jute	0.31	0.465	1.8	17.2	25.8	1.50
Ramie	0.59	0.885	3.7	14.6	21.9	1.50
<i>Natural polymer</i>						
Tenasco	0.27	0.405	16.9	6	9	1.50
Fortisan	0.59	0.885	6.4	16.1	24.15	1.50
<i>Synthetic</i>						
Nylon 6.6 HT	0.66	0.752	16	4.4	5	1.14
Polyester HT	0.56	0.778	7	13.2	18.35	1.39
Polypropylene	0.65	0.591	17	7.1	6.46	0.91
Para-aramid HM	2	2.88	2.4	80	115.2	1.44
Carbon HM	1.20	2.196	0.7–1.7	256	468.48	1.83
E-glass	0.78	2	4	28	72.24	2.58
Basalt CBF	0.7–0.9	1.8–2.5	3.1	30–35	80–93	2.65
PVA fibres	0.7–1.23	0.91–1.60	7	6.5–17	8.45–22	1.3
SWNTs	10–41	13–53	16	770–3846	1–5 × 10 ³	1.3–1.4
<i>Metals</i>						
Stainless steel fibres	0.22–0.28	1.7–2.21	1.5–11	25.19	199	7.9
Steel	–	0.34–2.1	–	–	210	7.8
Aluminium	–	0.14–0.62	–	–	70	2.7

HT: high tenacity

HM: high modulus

CBF: continuous basalt fibre

PVA: polyvinyl alcohol

SWNTs: Single-Walled carbon NanoTubes

Tex is the mass in g of 1 km of fibre or yarn

1.5 Natural textile fibres for use in civil engineering applications

1.5.1 Flax

Flax is a bast fibre and it comes from the stem of the *Linum usitatissimum* plant, which is an annual plant that grows in various temperate and sub-tropical regions around the world. The erect plant grows to about 1.2 m. The plant is also known as flax and the textiles made of it are called linen and are mainly used in apparel and home textiles.

The highest quality flaxes come from Belgium, Northern France and Holland. Russian flaxes are normally weaker but give very fine fibres.

Structure and properties

Flax is a unicellular fibre and its length may vary from 45–60 cm with a mean diameter of 0.02 mm. Under the microscope the cells look like long and transparent cylindrical tubes. The fibre cell has a narrow lumen running through the centre and the cell walls are thick and polygonal in cross-section.

Flax has an average tenacity of around 0.57 N/tex and an extension at break of around 1.8% (dry fibre) or 2.2% (wet fibre). It has a specific gravity of 1.54 and a regain of about 12%. It resists heat decomposition up to 120°C, when it begins to discolour. It is a good heat conductor. Under normal environmental circumstances flax has a high resistance to rotting. However, in a damp, warm and contaminated condition it may be attacked by mildews.⁵

1.5.2 Hemp

Hemp is a bast fibre and it comes from a plant known as *Cannabis sativa*, which is an annual plant cultivated in many countries around the world, the major producers being Canada, France and China. The plant may grow to a height of 3 m or even taller, and it is used mainly to make string, cord, rope, sacks and canvas.

Structure and properties

Hemp is dark in colour and the individual cells are cylindrical and thicker than flax with an average length of around 13–26 mm, presenting many irregularities on the surface. The cells of the hemp fibres have a thick wall and are polygonal in cross-section, and the central lumen is wider than that of flax. Hemp is stiffer than flax as it is more lignified. It has an average tenacity of around 0.47 N/tex and an extension at break of around 2.2%, with a specific gravity of 1.50.⁵

1.5.3 Jute

Jute is a bast fibre and it comes from the inner bark of an annual herbaceous plant known as *Corchorus*, which grows in the hot and damp regions of Asia, mainly in India and Bangladesh. The plant may grow to a height of 5 m with a stalk diameter of around 20 mm. It is used for sacks and packaging materials, particularly for the storage and transport of agricultural products.

Structure and properties

Jute's colour may vary from dirty grey to yellow and brown, and it has a silky luster. It consists of bundles of individual fibres that may reach 2 m in length. The individual cells of jute have an average length of 2–6 mm. The cell surface is generally smooth presenting some nodes and cross-markings, and the fibres are covered with a layer of woody material. The cross-section of the cells is polygonal with thick walls and a broad lumen of oval cross-section and irregular along the length. Due to the irregular thickness of the cell walls, jute is not as strong or durable as flax or hemp, with an average tenacity of around 0.31 N/tex and an extension at break of around 1.8%. It has a specific gravity of 1.50 and it is a very hygroscopic fibre with a regain of about 13.75%, absorbing up to 23% of water in humid conditions. Due to the protective effect of lignin it is quite resistant to rot.⁵

1.5.4 Ramie

Ramie is a bast fibre and it comes from a perennial plant known as *Boehmeria nivea*, which is grown mainly in China, South America, The Philippines, Korea, Japan and Indonesia. The plant sends up many stalks that may reach a height of 1.2–2 m. As a result of its high molecular crystallinity, ramie is stiff and brittle and breaks if flexed repeatedly in the same place. It lacks resiliency and is low in elasticity and elongation potential. Its main application is in heavy industrial fabrics, mainly canvas and packing materials.

Structure and properties

Ramie fibres are lustrous and white. The fibre is durable and stiff and it absorbs water quite readily. The individual cells are up to 45 cm in length and are smooth and cylindrical with thick walls. The surface of the cell has markings and the lumen disappears towards the ends of the cell, which tapers to a round point. The fibres are strong with an average tenacity of around 0.59 N/tex and an extension at break at around 3.7%, with a specific gravity of 1.50.⁵

1.6 Synthetic textile fibres for use in civil engineering applications

1.6.1 Nylon

Nylon is a generic name, given to a group of synthetic polymers known as polyamides, which was first produced in 1935 by Carothers at DuPont. Nylon was the first truly synthetic fibre to be commercialized (1939). It is a polyamide fibre, derived from a diamine and a dicarboxylic acid and, because a variety of diamines and dicarboxylic acids can be produced, there are a very large number of polyamide materials available to produce nylon fibres. The two most common versions are nylon 6/6 (polyhexamethylene adipamide) and nylon 6 (polycaprolactam, a cyclic nylon intermediate). Raw materials for these are variable and sources used commercially are benzene (from coke production or oil refining), furfural (from oat hulls or corn cobs) or 1,4-butadiene (from oil refining). There are other commercial nylon products, such as nylon 11, 12, 6/10, 6/12, and so on. These others are mainly used in tubing extrusion, injection moulding, and coatings of metal objects.⁶ Because nylons offer good mechanical and thermal properties, they are also a very important engineering thermoplastic. For example, 35% of total nylon produced is used in the automobile industry.⁷ Nylon's outstanding characteristic in the textile industry is its versatility. It can be made strong enough to stand up under the punishment tyre cords must endure, fine enough for sheer, high-fashion hosiery, and light enough for parachute cloth and backpackers' tents.⁸ Nylon is used both alone and in blends with other fibres, where its chief contributions are strength and abrasion resistance. Nylon washes easily, dries quickly, needs little pressing, and holds its shape well since it neither shrinks nor stretches. It is produced by melt spinning and is available in staple, tow, monofilament, and multi-filament form. The fibre has outstanding durability and excellent physical properties.

Structure and properties

Nylons are semi-crystalline polymers. The amide group – (–CO–NH–) – provides hydrogen bonding between polyamide chains, giving nylon high strength at high temperatures and toughness at low temperatures, combined with its other properties, such as stiffness, wear and abrasion resistance, low friction coefficient and good chemical resistance. These properties have made nylons one of the strongest man-made fibres in common use. Two-step melt spinning, comprised of spinning and drawing, is considered to be the conventional method to manufacture nylon filaments. After melting, filtering, and de-aerating, the molten polymer is extruded through a spinneret into a chamber where the melt solidifies into a filament form. These filaments must be drawn subsequently to increase molecular orientation and crystallinity, and therefore improved properties such as initial modulus, tenacity, elongation-at-break, density equilibrium and moisture sorption.

Rather than two-step spinning (extrusion) and drawing, a one-step, high-speed spinning process is being used increasingly.

Nylon 6/6 has a melting point of 263°C, high for a synthetic fibre, though not a match for aramids. This fact makes it the most resistant to heat and friction and enables it to withstand heat setting for twist retention. Nylon 6 has a much lower melting point than nylon 6/6.⁸

Nylon 6/6 has the following properties:

- tenacity (N/tex): around 0.66
- extension at break (%): in the region of 16
- elastic recovery at 8% extension (%): 100
- specific gravity: 1.14
- melting point: 263°C
- thermal conductivity ($\text{mW m}^{-1} \text{K}^{-1}$): 250
- extremely chemically stable
- no mildew or bacterial effects
- moisture regain (%): 4–4.5
- degraded by light as natural fibres
- permanent set by heat and steam
- abrasion resistant
- resilient.

1.6.2 Polyester

Polyesters are a type of polymer obtained by a condensation reaction between small molecules which link to form ester groups. There is a variety of polyesters, however, the most common one is polyethylene terephthalate (PET) which is obtained by the condensation of terephthalic acid or a derivative (e.g. dimethyl terephthalate) with ethylene glycol.

Polyesters can be made into fibres. The fibre-forming substance is any long-chain synthetic polymer composed of at least 85% by weight of an ester of a substituted aromatic carboxylic acid, including but not restricted to substituted terephthalic units, $\text{p}(-\text{R}-\text{O}-\text{CO}-\text{C}_6\text{H}_4-\text{CO}-\text{O}-)_x$ and parasubstituted hydroxybenzoate units, $\text{p}(-\text{R}-\text{O}-\text{CO}-\text{C}_6\text{H}_4-\text{O}-)_x$.

Polyesters may also be produced in numerous other forms such as sheets and three-dimensional shapes. While combustible at high temperatures, polyesters tend to shrink away from flames and self-extinguish upon ignition. Polyester fibres have high tenacity and E-modulus as well as low water absorption and minimal shrinkage in comparison with other fibres.

Structure and properties

The properties of polyester fibres are strongly affected by the fibre structure. The fibre structure depends considerably on the process parameters of fibre

formation, such as spinning speed, hot drawing, stress relaxation and heat-setting speed.

Any stretching of the fibre, whether obtained during fibre spinning or hot drawing, extends the PET molecules, resulting in better uniformity, lower elongation and higher strength, greater orientation, and high crystallinity. Relaxation consists of relieving the stresses and strains of the extended molecules, which results in a reduced shrinkage of the fibres. Heat setting is a treatment used to stabilize the molecular structure, enabling the fibres to resist dimensional changes. The degree of crystalline orientation and the percentage of crystallinity may be adjusted in response to process parameters and this strongly affects fibre properties.

The glass transition is usually around 75°C and crystallization and melting are around 130°C and 260°C, respectively.

Polyester fibres display good resistance to sunlight, but long-term degradation appears to be initiated by UV radiation. Although PET is flammable, the fabric usually melts and drops away instead of spreading the flame. Polyester fibres are resistant to mildew, aging and abrasion.⁹

The properties of PET may be summarized as:

- strong
- resistant to stretching and shrinking
- resistant to most chemicals
- quick drying
- crisp and resilient when wet or dry
- wrinkle resistant
- mildew resistant
- abrasion resistant
- thermal conductivity ($\text{mW m}^{-1} \text{K}^{-1}$): 140
- specific gravity: 1.38
- moisture regain (%): 0.4
- melting temperature (°C): 258–263.

In addition, polyester high-tenacity filament yarn has the following properties:

- tenacity (N/tex): 0.62–0.85
- extension at break (%): 10–20
- elastic recovery at 5% extension (%): 90
- initial modulus (N/tex): 10.2–10.6.

1.6.3 Polypropylene

Polypropylene (PP) is extremely versatile as a fibre-forming material. It is a thermoplastic polymer that may be used in a range of applications, including fibres, packaging, stationery, automotive components, and so on. Since its introduction into the textile industry in the 1950s, the list of successful products and markets for polypropylene fibre has increased exponentially.

Polypropylene may be recycled and its recycling ID code is 5. Melt processing of polypropylene may be achieved through extrusion and moulding, and the extrusion methods normally used include melt-blown and spun-bond fibres to be used in products such as face masks, filters, nappies and so on.

Structure and properties

Polypropylene fibres are composed of crystalline and non-crystalline regions. Fibre spinning and drawing causes the orientation of both crystalline and amorphous regions, and may lead to the formation of highly anisotropic microfibrillar structures, which in turn may result in anisotropic fibre properties.¹⁰

Polypropylene fibres have the following properties:

- very lightweight
- specific gravity: 0.91 (very low)
- high strength (wet or dry)
- tenacity (N/tex): 0.65
- extension at break (%): 17
- initial modulus (N/tex): 7.1
- thermal conductivity ($\text{mW m}^{-1} \text{K}^{-1}$): 120
- resistant to deterioration from chemicals, mildew, insects, perspiration, rot and weather
- abrasion resistant
- low moisture absorption
- stain and soil resistant
- lowest static component of any man-made fibre
- sunlight resistant
- good washability, quick drying, unique wicking
- resilient, mouldable, very comfortable
- thermally bondable.

1.6.4 Aramid

Aramid fibres are highly oriented aromatic polyamides that are heat resistant and very strong. The US Federal Trade Commission describes aramid fibre as ‘a manufactured fibre in which the fibre-forming substance is a long-chain synthetic polyamide in which at least 85% of the amide linkages, $(-\text{CO}-\text{NH}-)$, are attached directly to two aromatic rings’. The first meta-aramid fibre was introduced by DuPont (USA) in the early 1960s under the tradename of Nomex. Various other meta-aramids are: Teijinconex produced by Teijin (Japan), Kermel by Kermel (France) and New Star by Yantai (China). These have excellent heat resistance and they do not melt or ignite under normal levels of oxygen. These fibres are used in fire-fighting apparel and other items that require heat resistance, amongst which are special gloves and racing car drivers’ suits.

Another group of aramid fibres is the para-aramids such as Kevlar, which was first introduced by DuPont (USA) in 1973. Other para-aramids were introduced at a later stage, such as Twaron by Akzo (Netherlands) and Technora by Teijin (Japan). These fibres have a much higher modulus of elasticity and tensile strength and a lower elongation at break than the meta-aramid fibres. They are used in applications that require energy absorption, such as bullet-proof apparel and armor and the reinforcement of composite materials amongst others.

Structure and properties

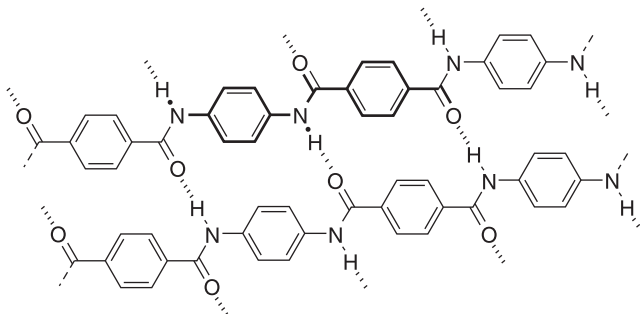
Rebouillat 11 presented a review of the relationship between structure and properties of aramid fibres.¹¹

Kevlar fibres are made up of long chains of molecules produced from PPTA (poly-paraphenylene terephthalamide), and inter-chain hydrogen bonds make these fibres extremely strong (Fig. 1.6).

The molecules that form the structure of Kevlar are relatively rigid, which is typical of planar sheet-like structures.

Aramid fibres have the following properties:

- high degree of orientation (which dominates their properties)
- good resistance to abrasion
- good resistance to organic solvents
- non-conductive
- no melting point, degradation starts from 500°C
- low flammability
- good fabric integrity at elevated temperatures
- sensitive to acids and salts
- sensitive to ultraviolet radiation
- prone to static build-up unless finished.



1.6 Molecular structure of Kevlar:²⁰ bold – monomer unit; dashed – hydrogen bonds.

In addition, para-aramids (such as Kevlar and Twaron) have the following properties:

- outstanding strength-to-weight properties
- tenacity (N/tex): 2 (high)
- extension at break (%): 2.4
- initial modulus (N/tex): 80 (high)
- low creep
- specific gravity: 1.44.

1.6.5 Carbon

Carbon fibres are mainly composed of carbon atoms that are bonded together to form microscopic crystals orientated along the fibre axis. These fibres are extremely fine with a diameter range of 0.005–0.010 mm. The fine filaments may be assembled into yarns and these may be woven into fabrics for the reinforcement of composite materials with a high strength-to-weight ratio, which is ideal for applications in aerospace, civil engineering, military, motor-sports and other competition sports.¹²

Rayon-based carbon fibres are only of historical interest. Most commercial production of carbon fibres is based on polyacrylonitrile (PAN) technologies. Pitch-based fibres are used for niche applications while vapor-grown fibres are at a starting stage.

Structure and properties

The physical properties of carbon fibres have been reviewed extensively by Dresselhaus *et al.*¹³

The characterization of carbon fibre microstructure has been mainly performed by X-ray scattering and electron microscopy techniques. In contrast to graphite, the structure of carbon fibre lacks any three-dimensional order. In PAN-based fibres, the linear chain structure is transformed to a planar structure during oxidative stabilization and subsequent carbonization. Basal planes oriented along the fibre axis are formed during the carbonization stage.

In general, it is seen that the higher the tensile strength of the precursor the higher is the tenacity of the carbon fibre. Tensile strength and modulus are significantly improved by carbonization under strain when moderate stabilization is used. X-ray and electron diffraction studies have shown that in high modulus type fibres, the crystallites are arranged around the longitudinal axis of the fibre, with layer planes highly oriented parallel to the axis. Overall, the strength of a carbon fibre depends on the type of precursor, the processing conditions, heat treatment temperature and the presence of flaws and defects. Carbon fibres are very brittle because the layers in the fibres are formed by strong covalent bonds.

The sheet-like aggregations allow easy crack propagation and, on bending, the fibre fails at very low strain.¹⁴

Carbon HM fibres have the following properties:

- specific gravity: 1.83
- tenacity (N/tex): 1.2 (high)
- extension at break (%): 0.7–1.7
- initial modulus (N/tex): 256 (high).

1.6.6 Glass

Fibreglass is made up of very fine fibres of glass. It has a number of applications mainly for insulation, filtration media, reinforcement, and as optical fibres. It is quite an ancient technology but the great push was given by Russell Games Slayter of Owens-Corning in 1938. The basis of textile-grade glass fibres is silica (SiO_2) and in its pure form it exists as a polymer $(\text{SiO}_2)_n$.

Structure and properties

Fibreglass or glass fibre, as it is also known, has no true melting point. However, it softens at 2000°C, where it starts to degrade. At 1713°C most of the molecules can move about freely, and if the glass is then cooled quickly they will be unable to form an ordered structure.¹⁵ In the polymer, it forms SiO_4 groups that are configured as a tetrahedron with the silicon atom at the centre and four oxygen atoms at the corners. These atoms then form a network bonded at the corners by sharing the oxygen atoms. Glass fibre is formed when thin strands of silica-based, or other formulation glass, are extruded into many fibres with small diameters suitable for textile processing. Glass, even as a fibre, has little crystalline structure. The properties of the structure of glass in its softened stage are very much like its properties when spun into fibre. One definition of glass is ‘an inorganic substance in a condition which is continuous with, and analogous to, the liquid state of that substance, but which, as a result of a reversible change in viscosity during cooling, has attained so high a degree of viscosity as to be, for all practical purposes, rigid’.¹⁶

By trapping air within them, blocks of glass fibre make good thermal insulation, with a thermal conductivity in the order of 0.05 W/mK. Because glass has an amorphous structure, its properties are the same along the fibre and across the fibre. Humidity is an important factor in the tensile strength.

E-Glass fibres have the following properties:

- specific gravity: 2.58
- tenacity (N/tex): 0.78
- extension at break (%): 4
- initial modulus (N/tex): 28
- specific heat ($\text{J g}^{-1}\text{K}^{-1}$): 0.8.

1.6.7 Basalt

The basalt fibre is made up of very fine fibres of basalt, and its composition consists of the minerals plagioclase, pyroxene and olivine. It has many of the applications of carbon and glass fibre. It is used as a textile for fireproofing in aerospace and automotive applications, and it can also be used as a composite reinforcement, including concrete reinforcement, in the construction of buildings, roads and runways.

Structure and properties

Basalt is a common extrusive volcanic rock, which is usually grey to black and fine-grained due to rapid cooling of lava at the surface of a planet. It may be porphyritic, containing larger crystals in a fine matrix, or vesicular, or frothy scoria. Unweathered basalt is black or grey. The manufacture of basalt fibre requires the melting of the quarried basalt rock to about 1400°C, this is then extruded through small nozzles to produce continuous filaments of basalt fibre. There are three main manufacturing techniques, which are centrifugal-blowing, centrifugal-multiroll and die-blowing. The fibres typically have a filament diameter of 9–13 µm, which is far enough above the respiratory limit of 5 µm to make basalt fibre a suitable replacement for asbestos. They also have a high elastic modulus, resulting in an excellent specific tenacity that is three times that of steel. The continuous basalt fibres derived from basalt rock have proven technical characteristics and performance specifications.

Basalt fibres have the following properties:^{17,18}

- fibre diameter (µm): 9
- specific gravity: 2.65
- high thermal resistance (thermostability) and low flammability
- low-strength degradation at temperatures as low as –200 to 250°C and as high as +700 to 900°C., and of high humidity
- operative temperature (°C): –200 to +900
- high thermal and acoustic insulation properties
- sound proofing for 400–1800 Hz: 80–95%
- excellent adhesion to polymer resins and rubbers
- relatively high mechanical strength, abrasion resistance and elasticity
- tenacity (N/tex): 0.67–0.93
- extension at break (%): 3.1
- initial modulus (N/tex): 30–35
- high dielectric properties
- moisture regain (%): 1
- low water absorption
- high chemical resistance (especially to concentrated acids-based materials)
- ecologically clean and non-toxic.

1.6.8 Polyvinyl alcohol (PVA) fibres

PVA fibres are essentially composed of polyvinyl alcohol and produced by wet spinning, and also by a special process that combines dry and melt spinning. There are various types of PVA fibres. However, in this section only the high-strength and high-modulus PVA fibres will be considered. They are high strength, high modulus, acid-proof, alkali-proof, abrasion resistant and sunlight resistant, and these properties render them excellent for use as fabric in textiles, as a reinforcing material in building construction applications, as a packaging material, and in ropes and meshes. PVA fibres are high-performance fibres for use in cement, concrete and mortar, and they can be used together with steel reinforcement, where PVA fibres' micro-crack control helps to prevent water penetration and improves toughness and stiffness. They can also be used on their own as a truly structural fibre. Like steel, carbon, aramid and fibreglass, these PVA fibres have higher tensile strength and a greater modulus of elasticity than regular concrete. Unlike steel and other structural fibres, PVA is hydrophilic, causing it to form a molecular bond with the cement matrix during hydration and curing. This unique combination of bond strength, tensile strength and high modulus of elasticity gives PVA-reinforced concrete substantial tensile and flexural strength. In an engineered matrix, PVA fibres may be able to absorb as much as 500 times more energy than typical fibre-reinforced concrete in tension and bending. PVA fibres are produced in a variety of diameters including the range from 0.027 to 0.7 mm, and in a variety of cut lengths including the range from 6 to 30 mm, both depending on the application envisaged.

In PVA-engineered cementitious composites (PVA-ECC) a small percentage (around 1%) of PVA fibres are effective in controlling:

- plastic shrinkage cracking
- drying shrinkage cracking
- autogenous cracking
- fatigue cracking
- spalling.¹⁹

Structure and properties

The PVA molecule contains a great number of hydroxyl groups and polyvinyl alcohol itself is soluble in water, and so the initially developed fibres were limited to specialized water-soluble fibre applications. Efforts were made in Japan to produce water-resistant PVA fibres that succeeded and resulted in the high-strength/high-modulus PVA fibres used in civil engineering applications. The latter fibres are heat treated or acetalized to render them insoluble in water.

Fibres are smooth on the surface, and white with silk-like lustre. The cross-section is generally U-shaped like a flattened tube. There is a pronounced skin layer, which is more crystalline than the core. The mean value of crystallinity is around 50%.²⁰

PVA fibres have the following properties:

- specific gravity: 1.3
- tenacity (N/tex): 0.7–1.23
- extension at break (%): 7%
- initial modulus (N/tex): 6.5–17.

1.6.9 Metallic fibres

The ductility of metals makes possible their conversion into wire by drawing the metal through dies that become successively smaller. They are quite rigid materials and this rigidity imposes mechanical difficulties in textile processing. However, the flexibility of materials is a function of the cross-sectional area. As the diameter of a rod is reduced by one-half, its flexibility is increased by a factor of four. The problem of producing metal fibres of adequate flexibility resolves itself, therefore, into that of producing filaments of the necessary fineness. In the case of a metal such as stainless steel, a filament of approximately $12\mu\text{m}$ has a flexibility equivalent to that of a nylon filament of 3.3 dtex. During the 1960s, methods of producing metal filaments of the required fineness were developed essentially for military and aerospace applications because of the high cost and low production quantities.

Stainless steel fibre is a grey filament of near-round cross-section with diameters that can vary from 4 to $50\mu\text{m}$.²⁰

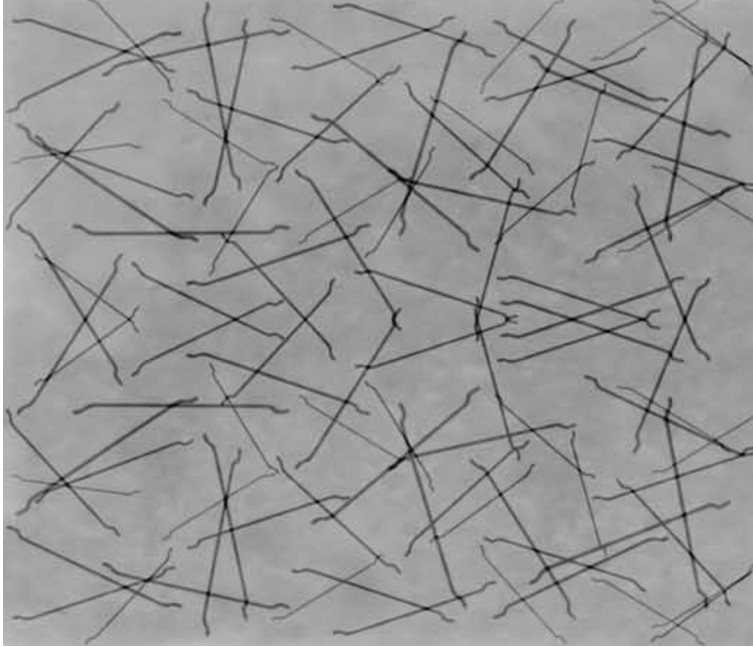
Subsequent developments enabled metal fibres to be produced in large quantities and at a lower cost, making them competitive for application in other areas.

The most important metallic fibre for civil engineering applications is steel and so this section will be dedicated to this type of metal fibre.

Concrete reinforced with steel fibre as compared to hand-tied rebar is less expensive and still increases the tensile strength quite considerably. The shape and dimensions of the fibre are important aspects to consider. For shotcrete a standard size fibre would be about 1 mm in diameter and 45 mm in length, but for steel fibres other lengths may be used (30 to 80 mm) and shapes (i.e. end-hooks). Steel fibres should only be used on surfaces that are corrosion and rust-stains free. Some steel fibre surfaces may be covered by special materials (Fig. 1.7).



1.7 Different types of steel fibres used for concrete reinforcement.



1.8 Steel fibre-reinforced concrete.¹³

Steel fibre-reinforcement offers constructors an efficient and cost-effective alternative for mesh or rebar reinforcement (Fig. 1.8). Working with fibre reinforcement not only means using less steel, but it also takes less man hours to put in place. Steel fibre reinforcement also offers ultimate flexibility. Even complex shapes and structures are possible. Steel fibres have the ability to make concrete more ductile, and steel fibres, having hooked ends, enable the best anchorage and controlled pull-out.²¹

Structure and properties

The properties of a metal fibre are essentially those of the metal itself.

The molecular architecture of metals imposes a characteristic behaviour under stress that differs from that of a typical organic textile fibre. The stress–strain curve of a metal fibre shows a completely elastic behaviour up to the yield point. It is devoid of viscous components such as are found in organic high polymers, and which are identified as primary creep. The absence of a significant viscous behaviour characteristic of metal fibres ensures dimensional stability of structures.

A typical cold-drawn steel wire fibre with hooked ends used for concrete reinforcement may have a diameter of 0.8 mm, a length of 60 mm and a tensile strength of 1000 N/mm².

Stainless steel fibres (type 304) have the following properties:

- specific gravity: 7.9
- tenacity (cN/tex): 22.1–28.3
- extension at break (%): 1.5 (hard)–11 (annealed)
- modulus of elasticity (N/tex): 25.19.

1.6.10 Nanofibres

Polymeric nanofibres may range in diameter from 50 to 500 nanometers, since a polyester crystallite has dimensions in the order of 40 nanometers. Polymeric nanofibres may be produced by several methods, including electrospinning, melt blowing a fibre with a modular die, and spinning bicomponent fibres that split or dissolve (e.g. islands-in-the-sea (INS) fibres using a standard spin/draw process).²²

Carbon nanotubes are an ordered array of carbon atoms which can have tensile strength up to 15 times that of steel. These tubes or fibres are often called graphite or carbon nanofibres as well as nanotubes. The technology for manufacturing carbon nanotubes is very different from common fibre-production techniques. Shorter nanotubes (0.05–3 μm range) may be produced by a variety of techniques including arc-discharge, laser oven, high-pressure CO conversion (HiPco) and fluidized bed chemical vapor deposition (CVD). Longer nanotubes (from 100 μm up to the cm scale) may be produced by substrate growth CVD and catalytic gas flow CVD.

Fibres containing only carbon nanotubes (CNTs) will yield great advances in high-tech applications if they can attain a significant portion of the extraordinary mechanical and electrical properties of individual CNTs. Doing so will require that the CNTs in the fibre are sufficiently long, highly aligned and packed in an arrangement that is nearly free of defects.

The method of CNT production strongly influences the type (multi-walled CNT vs. single-walled CNT), length (submicron to millimetre), chirality and processability, which in turn determine the properties of the final CNT-based macroscopic product.²³

An important advantage of nanofibres is the very high increase in surface area to volume ratio leading to a higher specific surface area. This may lead to a considerable enhancement of all sorts of fibre properties. In relation to strength, it was recognized by Griffiths that the strength of any material was determined by the presence of flaws of critical dimension. He demonstrated for glass that increasing the surface-to-volume ratio would lead to increased strength by reducing the number of flaws of critical dimension. Thus, high-strength materials will be filamentary of fine diameter. In this context the strength of nanofibres should be extremely high.²⁴

Thanks to their extraordinary properties, some of the nanofibres described, especially CNTs, may be of great interest in civil engineering applications in the future.

Structure and properties

Carbon nanotubes have attracted a lot of attention in recent years for their extraordinary properties. They are cylindrical nanostructures of allotropes of carbon, and some of the most recently developed nanotubes may have a length-to-diameter ratio of up to 132 000 000:1.

Because of their properties, they find applications in a variety of fields including nanotechnology, electronics, optics and potentially in architectural fields. They are extremely strong, have very interesting electrical properties and are excellent thermal conductors.

Nanotubes belong to the fullerene structural family. Nanotube chemical bonding may be described by orbital hybridization, which is a field of applied quantum chemistry, and it is composed entirely of sp^2 bonds, which are identical to those of graphite. Another interesting characteristic of carbon nanotubes is that they self-align, forming 'ropes' held together by Van der Waals forces.

Single-walled CNTs have the following properties:

- specific gravity: 1.3–1.4
- tenacity (N/tex): 10–41
- extension at break (%): 16
- modulus of elasticity (N/tex): 770–3846.

1.7 Fibre–matrix adhesion

It is very important that there is good adhesion between the reinforcing fibres and the concrete or cement matrix, otherwise debonding may take place. Because fibres and cement are different materials, poor adhesion at the interface is a problem that often occurs. To improve adhesion, various approaches may be used, including the increase of the specific surface area of the fibre by changing the fibre cross-section from circular to irregular (e.g. serrated), coating the fibre with a material that promotes adhesion, using chemical agents to improve bonding, and so on.

The addition of latex to CFRC (carbon fibre-reinforced cement), and hot water curing of CFRC, dramatically increases fibre–matrix adhesion. Both latex (with an anti-foam agent) and hot water curing may increase the flexural strength by 40% over normal CFRC, and latex increases toughness 100%. Alteration of fibre–matrix adhesion changes the failure mode from fibre pull-out to fibre rupture. Optimum strength and toughness of CFRC result from an intermediate level of fibre–matrix adhesion.²⁵

1.8 Sources of further information and advice

The present chapter has the objective of presenting in an integrated and concise form the main fibres that are of use for civil engineering applications. For further

information the reader should consult more specialized literature, like those suggested below:

- Boron Fibre and SCS Silicon Carbide Fibre: <http://www.specmaterials.com>
- Composites in Civil Engineering: <http://www.tifac.org.in/news/civil.htm>
- Cook, J.C. (1984) *Handbook of Textile Fibres: Man-made Fibres*. Darlington: Merrow Pub. Co. Ltd.
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- International Jute Study Group: <http://www.jute.org/home2.htm>
- Large Volume, High Performance Application of Fibres in Civil Engineering: http://www.engineeredcomposites.com/publications/li_APS02.pdf
- Netcomposites: <http://www.netcomposites.com>
- Publications on Natural Fibre Composites: <http://www.agrofibrecomposites.com/publications.htm#26>
- The National Academies Press: <http://www.nap.edu/books/0309096146/html>

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